Course Registration System

Project plan, Use Case, BPM and ERD models

System Analysis and Design | Mini Project



Auralis



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PROJECT PLAN

Problem, objectives and business rules

The problem

Course registration puts heavy pressure on university staff at the start of every semester. When it is handled with paper forms or separate spreadsheets, each request is checked by hand (prerequisites, free seats, timetable conflicts), so errors and delays are common and managers cannot see current enrollment.

Objectives

- Analyze the users, main processes and business rules of course registration.
- Model the processes and requirements with BPM and Use Case diagrams.
- Design the data with Chen and IE ERDs and database tables.
- Show the main screens and keep all models consistent.

Key business rules

BR-1 A student can register only if the prerequisite course is passed.

BR-2 A section cannot exceed its capacity; a full section offers the waitlist.

BR-3 A student cannot register for two classes at the same time.

BR-4 A course can be dropped only before the drop deadline.

PROJECT PLAN

Scope, users and timeline

Scope

User accounts and login
Role-based access for students, staff, managers and administrators

Course and section management
Courses, sections, instructors, rooms, times and seat capacity

Registration and reports
Register, drop and waitlist with automatic checks; enrollment reports

Out of scope: Tuition payment, grades and transcripts, and software implementation. This project covers system analysis and design only.

Student
Searches courses, registers and drops, checks status

Staff
Registrar and instructors: manage sections and class lists

Manager
Department head: monitors seats and enrollment

System Administrator
Manages user accounts and access

Timeline

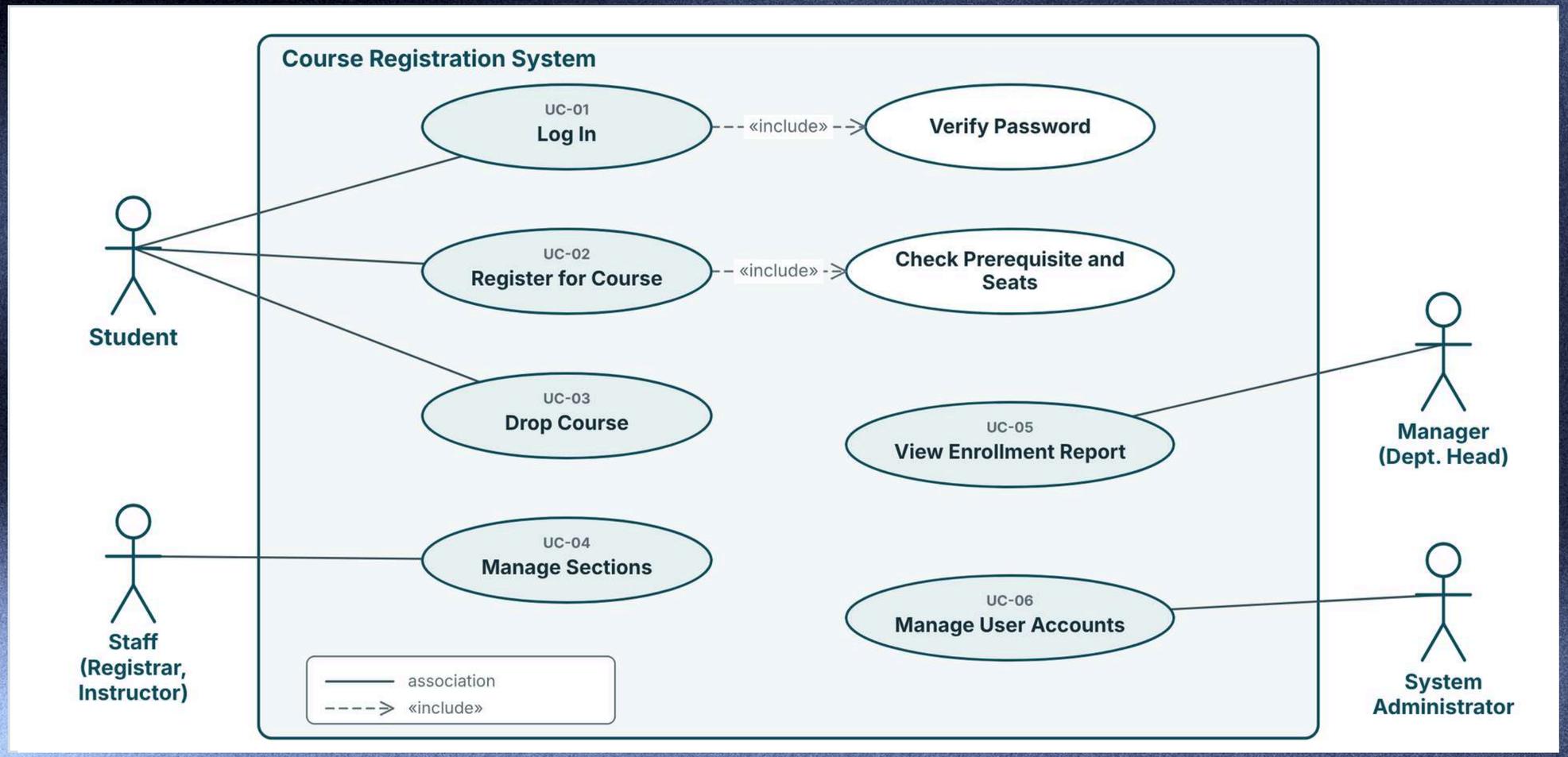
Week 1 | 5 - 11 Oct
Project plan and requirements

Week 2 | 12 - 18 Oct
Business process models

Week 3 | 19 - 25 Oct
Use case diagram and specifications

Week 4 | 26 Oct - 1 Nov
ERDs, database tables and review

Use Case Diagram: Course Registration System



Explanation Four actors use six use cases. The Student logs in (including password verification), registers for a course (including a check of the prerequisite and seats) and drops a course. Staff manage sections, the Manager views enrollment reports and the System Administrator manages user accounts.

Use Case Specifications (UC-02 to UC-05)

UC-02 Register for Course — Student

Precondition: The student is logged in and the registration period is open.
Main flow: 1. The student searches and selects a section. 2. The system checks the prerequisite. 3. The system checks that a seat is free. 4. The system checks the time conflict. 5. The system creates the registration, adds 1 to seats_taken and shows a confirmation.
Alternative: Prerequisite missing or time conflict: the request is rejected with the reason. Section full: the system offers the waitlist.
Postcondition: A REGISTRATION record exists with status Registered.

UC-03 Drop Course — Student

Precondition: The student has a registration with status Registered or Waitlisted.
Main flow: 1. The student selects the course and confirms the drop. 2. The system checks the drop deadline. 3. The system sets the status to Dropped and releases the seat. 4. The system promotes the first waitlisted student, if any. 5. The system shows a confirmation.
Alternative: Deadline passed: the drop is refused and the student is told why.
Postcondition: The registration status is Dropped and the seat is free again.

UC-04 Manage Sections — Staff (Registrar, Instructor)

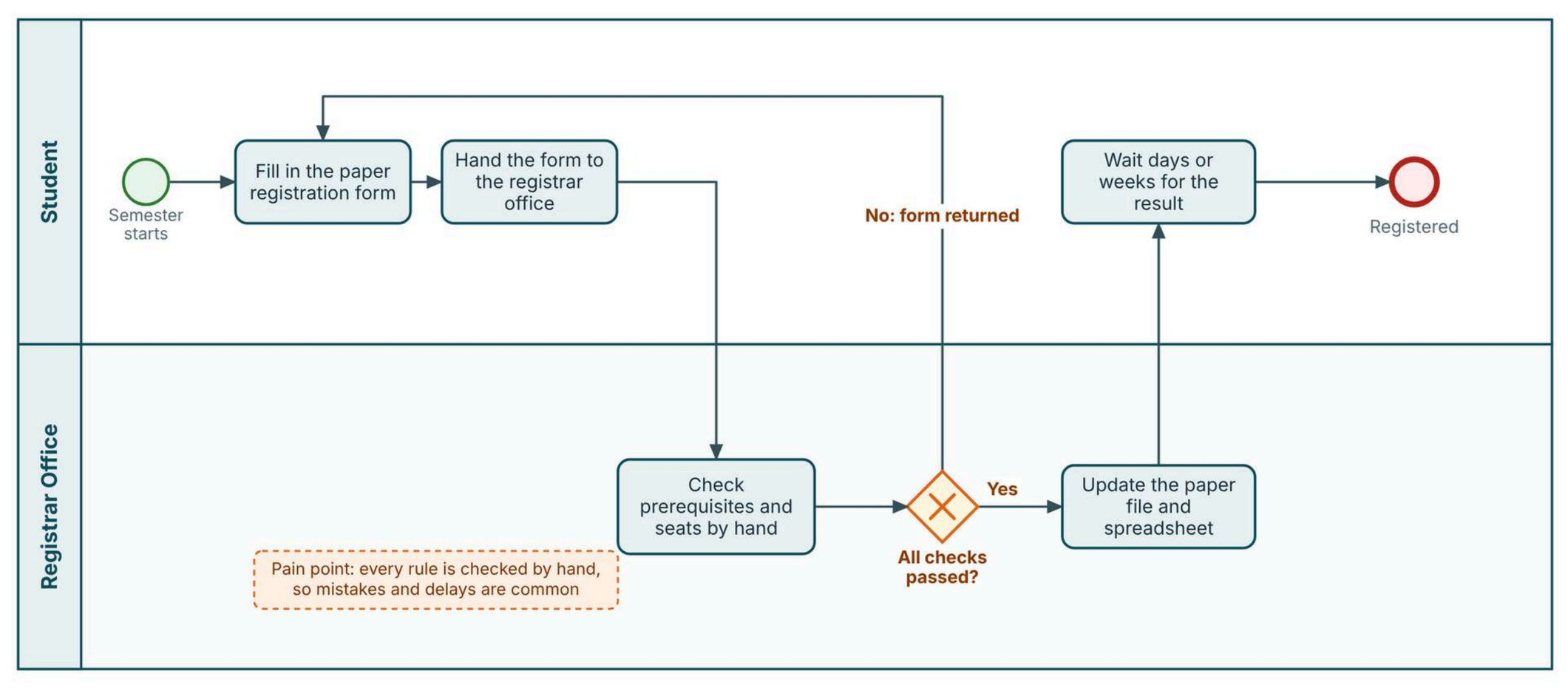
Precondition: The staff member is logged in.
Main flow: 1. Staff open the Manage Sections screen. 2. Staff create or edit a section: course, day and time, room, capacity and instructor. 3. The system checks room and instructor conflicts. 4. The system saves the section.
Alternative: Conflict found: the system shows a message and staff change the time or room.
Postcondition: The SECTION record is saved and students can register for it.

UC-05 View Enrollment Report — Manager (Department Head)

Precondition: The manager is logged in.
Main flow: 1. The manager opens the report page and selects a semester. 2. The system reads capacity and seats_taken of every section. 3. The system shows seats used, full sections and waitlisted students.
Alternative: No data for the semester: the system shows an empty report message.
Postcondition: The report is displayed; no data is changed.

Explanation Each specification lists the actor, the conditions before and after, the main step-by-step flow and what happens when a check fails. They describe the use cases in the diagram that are most important to students, staff and managers.

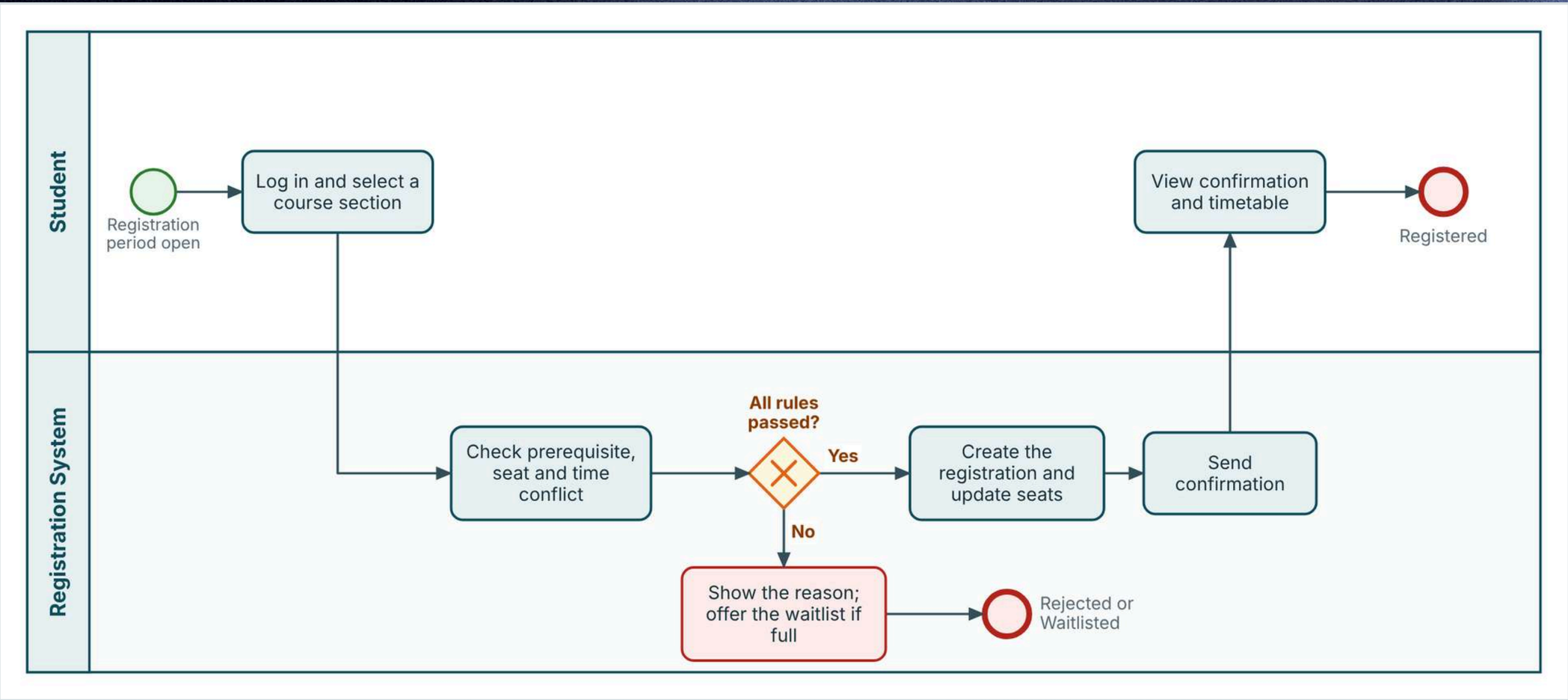
BPM 1: Current Registration Process (As-Is)



Explanation Today the student fills in a paper form and hands it to the registrar office. Staff check prerequisites and seats by hand, return the form if a check fails and update paper files. The student waits days or weeks for the result.

BUSINESS PROCESS MODELING

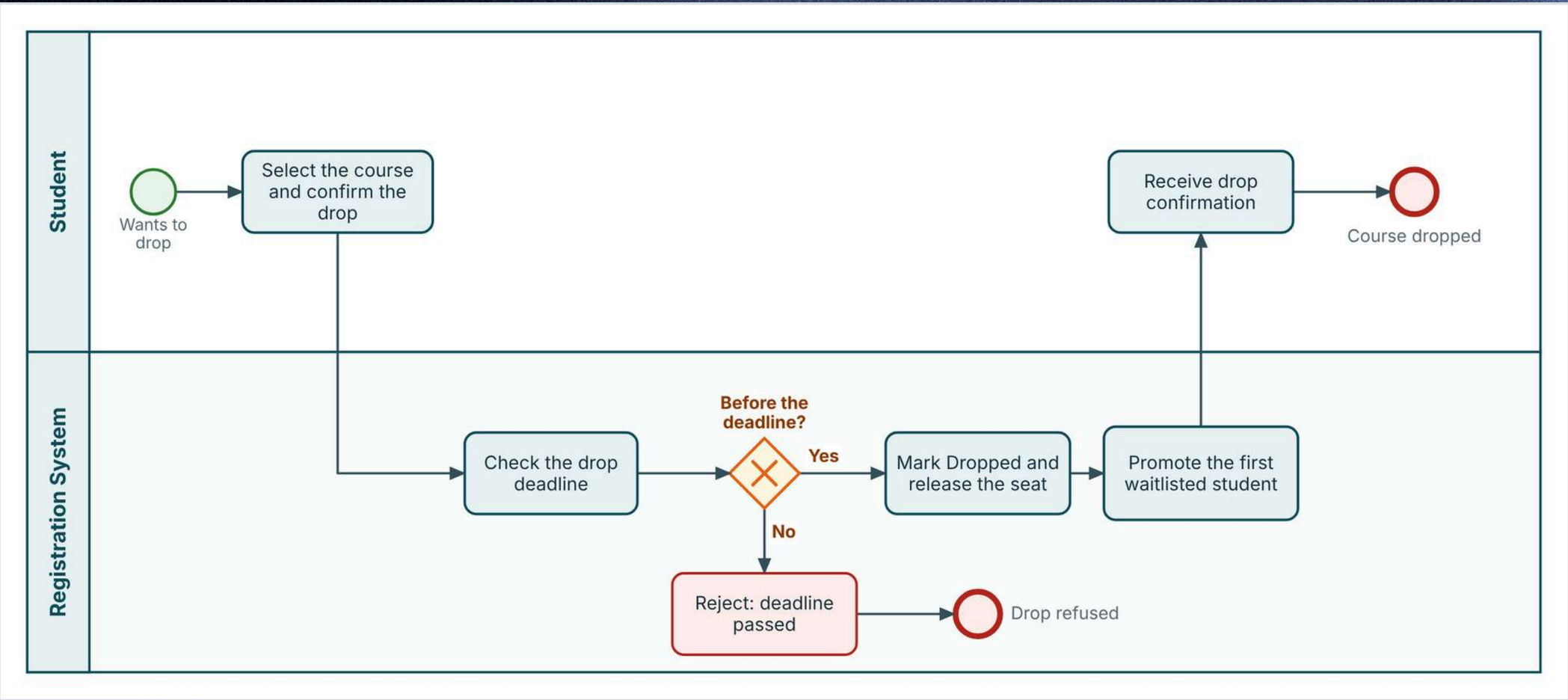
BPM 2: Register for a Course (To-Be)



Explanation In the proposed system the student selects a section and the system checks the prerequisite, seat and time conflict at once. If all rules pass it creates the registration, updates the seats and confirms; otherwise it shows the reason or offers the waitlist.

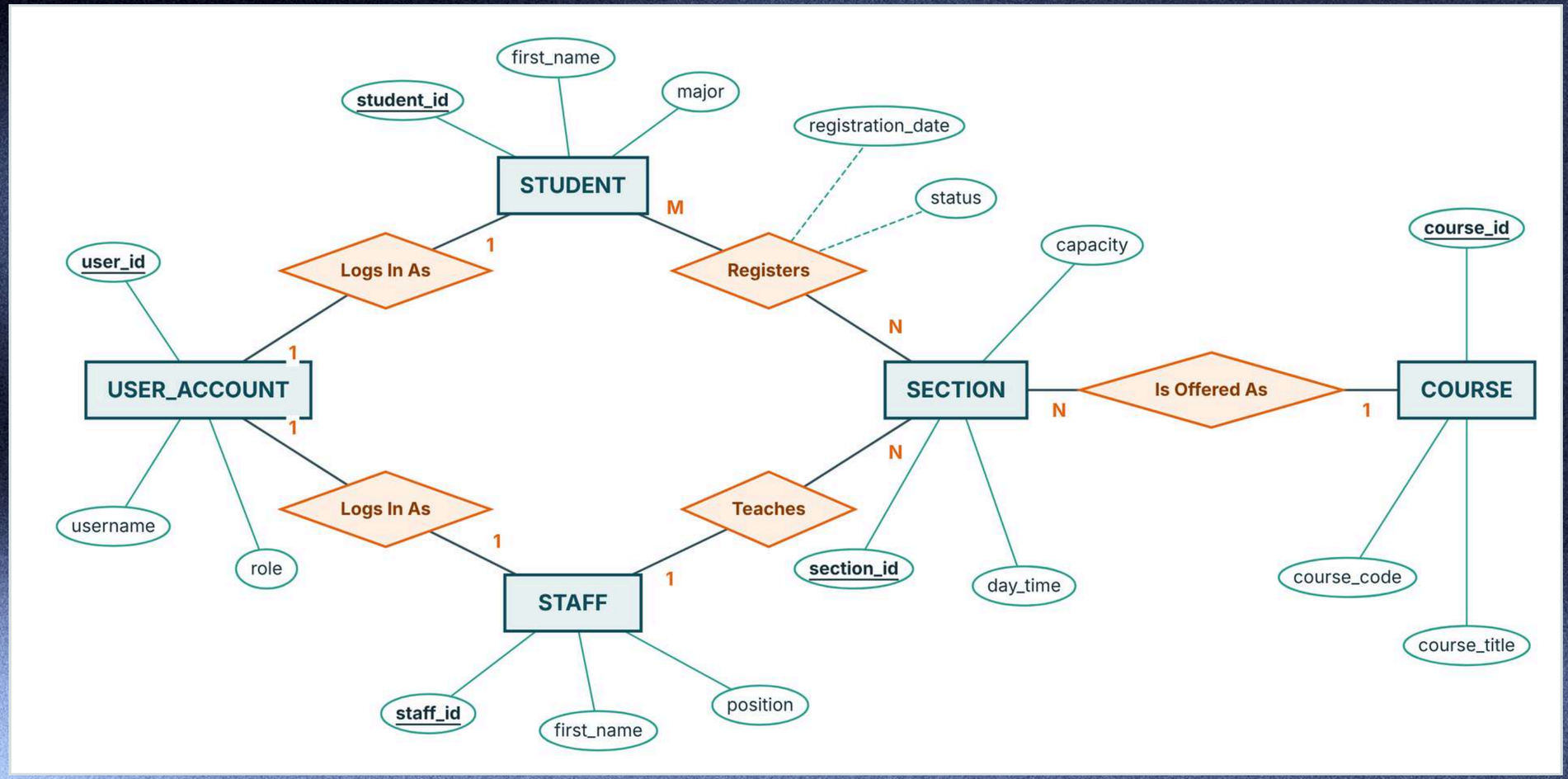
BUSINESS PROCESS MODELING

BPM 3: Drop Course



Explanation The system checks the drop deadline. Before the deadline it marks the registration Dropped, releases the seat and promotes the first waitlisted student; after the deadline the drop is refused. The student then receives a confirmation.

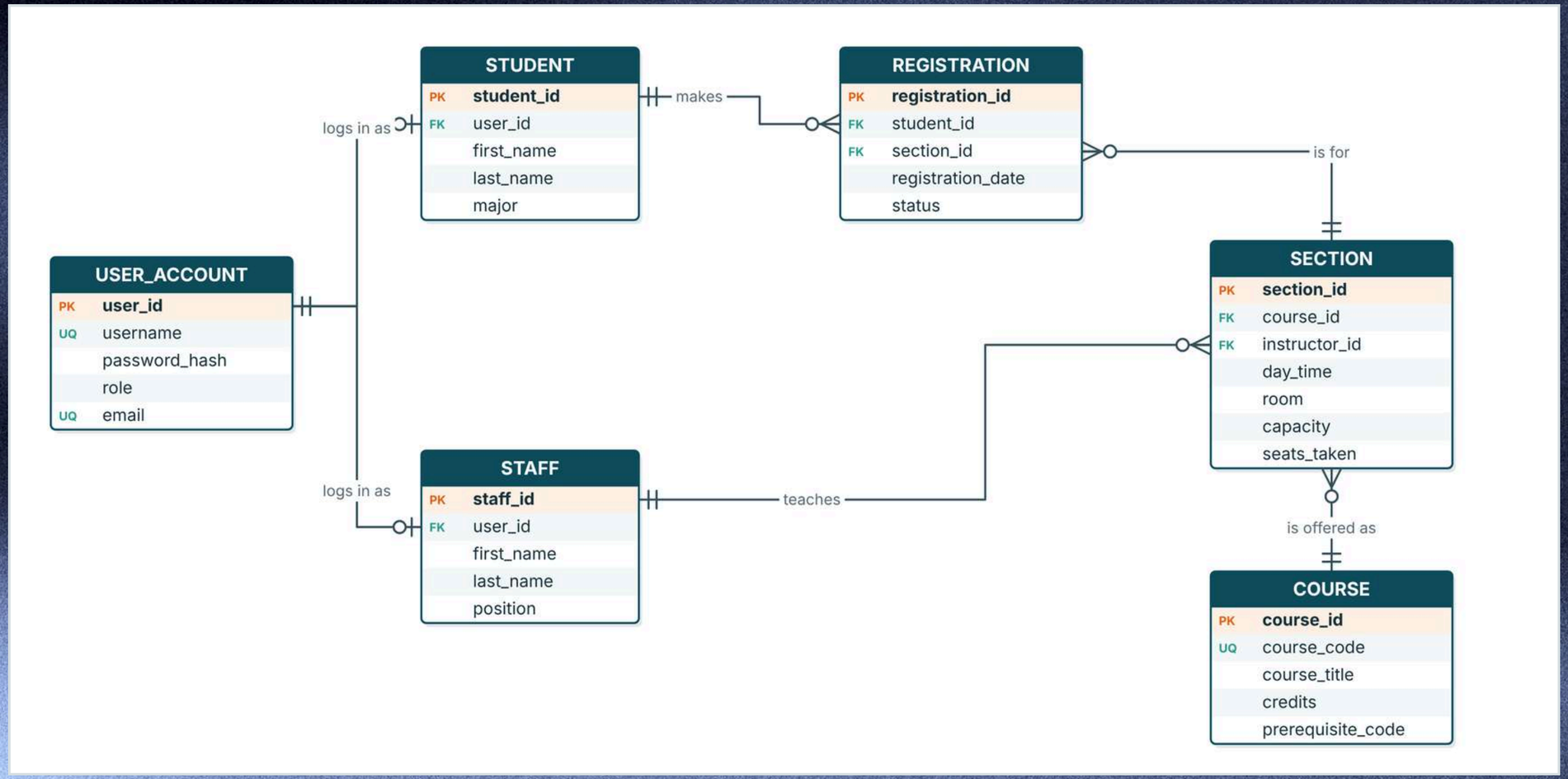
Chen ERD



Explanation A USER_ACCOUNT logs in as one STUDENT or one STAFF member (1:1). A STUDENT registers for many SECTIONs and a SECTION has many students (M:N); the relationship carries registration_date and status. A STAFF member teaches many SECTIONs, and a COURSE is offered as many SECTIONs (1:N). Underlined attributes are primary keys; only the main attributes are shown.

IE ERD

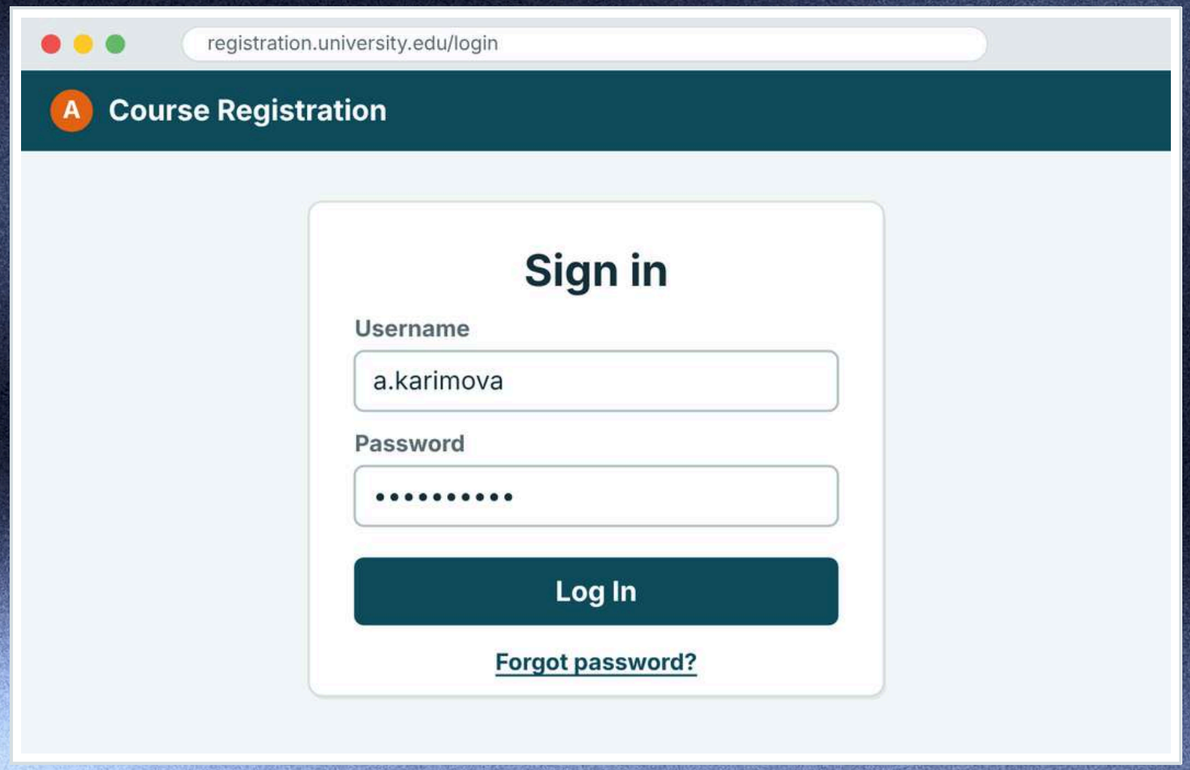
IE (Crow’s Foot) ERD



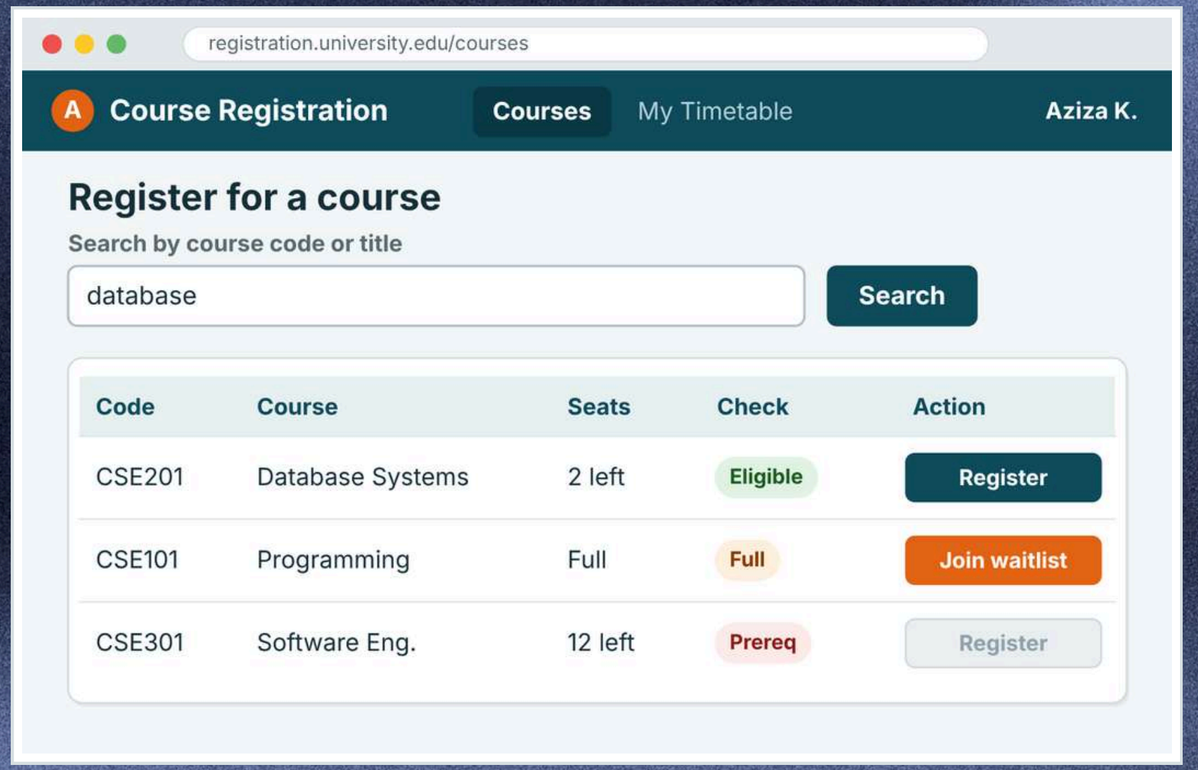
Explanation The IE ERD turns the Chen model into six tables. REGISTRATION resolves the many-to-many link between STUDENT and SECTION, and SECTION holds foreign keys to COURSE and to STAFF (the instructor). PK = primary key, FK = foreign key, UQ = unique; the crow’s foot marks the “many” side.

SCREEN LAYOUTS

Screen Layouts: Login and Course Registration



SCR-1 Login (UC-01)

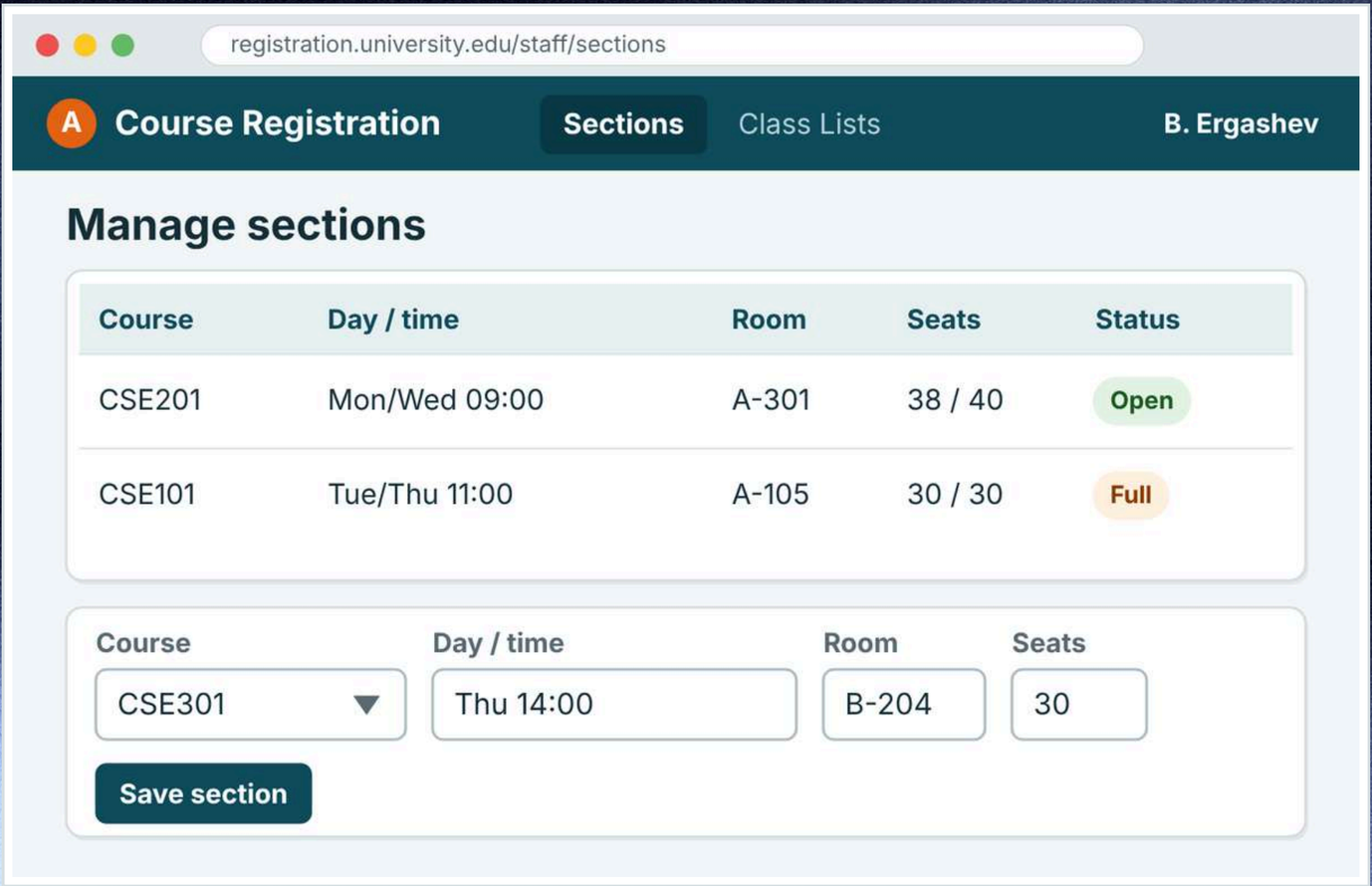


SCR-2 Register for a Course (UC-02)

Explanation SCR-1 is the login page used by every role. SCR-2 lists sections with the result of the automatic check (Eligible, Full or Prerequisite missing); the action button changes to Register, Join waitlist or a disabled button.

SCREEN LAYOUTS

Screen Layout: Section Management for Staff



Screens and use cases

SCR-1 Login
UC-01 Log In

SCR-2 Register for a Course
UC-02 Register for Course

SCR-3 Manage Sections
UC-04 Manage Sections

UC-03, UC-05 and UC-06 use the same layout style and are not drawn separately in this mini project.

Explanation SCR-3 lets staff view the sections with their time, room and seats, and add a new section with its course, day and time, room and capacity. The system checks conflicts when the section is saved.

DATABASE TABLES

Database Tables 1: USER_ACCOUNT, STUDENT and STAFF

USER_ACCOUNT — Login account of every user

PK user_id INT	UQ username VARCHAR(30)	password_hash VARCHAR(255)	role VARCHAR(15)	UQ email VARCHAR(80)
1	a.karimova	$2b$12$K9x...	Student	a.karimova@uni.edu
2	b.ergashev	$2b$12$Q4m...	Staff	b.ergashev@uni.edu

STUDENT — Student profile

PK student_id INT	FK user_id INT	first_name VARCHAR(40)	last_name VARCHAR(40)	major VARCHAR(50)
101	1	Aziza	Karimova	Computer Science
102	3	Jasur	Rahimov	Business

STAFF — Registrars, instructors and department heads

PK staff_id INT	FK user_id INT	first_name VARCHAR(40)	last_name VARCHAR(40)	position VARCHAR(30)
1	2	Bobur	Ergashev	Instructor
2	4	Dilnoza	Saidova	Registrar

Explanation These tables store who can log in and the profile of each student and staff member. Each STUDENT and STAFF row points to one USER_ACCOUNT through a foreign key (FK). PK = primary key, UQ = unique; two sample rows are shown for each table.

DATABASE TABLES

Database Tables 2: COURSE, SECTION and REGISTRATION

COURSE — Course catalog

PK course_id INT	UQ course_code VARCHAR(10)	course_title VARCHAR(80)	credits TINYINT	prerequisite_code VARCHAR(10)
1	CSE101	Programming Basics	3	*NULL*
2	CSE201	Database Systems	3	CSE101

SECTION — A course offered in one semester

PK section_id INT	FK course_id INT	FK instructor_id INT	day_time VARCHAR(30)	room VARCHAR(10)	capacity INT	seats_taken INT
1	2	1	Mon/Wed 09:00-10:30	A-301	40	38
2	1	1	Tue/Thu 11:00-12:30	A-105	30	30

REGISTRATION — A student's enrollment in a section

PK registration_id INT	FK student_id INT	FK section_id INT	registration_date DATE	status VARCHAR(12)
1	101	1	2026-09-01	Registered
2	102	2	2026-09-01	Waitlisted

Explanation COURSE is the catalog, SECTION is a course offered with its time, room, capacity and used seats, and REGISTRATION links a student to a section with a status (Registered, Waitlisted or Dropped). NULL means that the course has no prerequisite.

SUMMARY

How the models fit together

Use case	BPM	Main tables	Screen
UC-01 Log In	-	USER_ACCOUNT	SCR-1
UC-02 Register for Course	BPM-2	REGISTRATION, SECTION, COURSE, STUDENT	SCR-2
UC-03 Drop Course	BPM-3	REGISTRATION, SECTION	-
UC-04 Manage Sections	-	SECTION, COURSE, STAFF	SCR-3
UC-05 View Enrollment Report	-	SECTION, REGISTRATION	-
UC-06 Manage User Accounts	-	USER_ACCOUNT, STUDENT, STAFF	-

The project analyzed how course registration works today and modeled the proposed system with a use case diagram, three business process models, a Chen and an IE ERD, three screen layouts and six database tables. Names of actors, use cases and tables are the same in every diagram, so the models agree with each other and can be used as the base for later development.

Thank you | Auralis Team